

2012-2013 – Master 2 – Macro I

Lecture notes #11 : Competitive search equilibrium

Franck Portier
(based on Gilles Saint-Paul lecture notes)
franck.portier@TSE-fr.eu

Toulouse School of Economics

Version 1.1

19/11/2012

Changes from version 1.0 are in red

Disclaimer

These are the slides I am using in class. They are not self-contained, do not always constitute original material and do contain some “cut and paste” pieces from various sources that I am not always explicitly referring to (not on purpose but because it takes time). Therefore, they are not intended to be used outside of the course or to be distributed. Thank you for signalling me typos or mistakes at franck.portier@TSE-fr.eu.

Introduction

- ▶ The idea behind competitive search equilibrium is that search intermediaries could arise and internalize the congestion externalities and appropriability problems associated with search.
- ▶ This is similar to the issues arising in local public good provision and coasian internalization of externalities through contracts and property rights.

1. Fishing Clubs

Model

- ▶ a pond where the number of fish being caught is $F(N)$, where N is the number of fishermen.
- ▶ Each fisherman catches $F(N)/N$ fish
- ▶ The value of each fish is 1 and the opportunity cost of fishing per fisherman is c .

1. Fishing Clubs

Equilibrium without clubs and Optimum

- ▶ The equilibrium number of fishermen is such that

$$\frac{F(N)}{N} = c.$$

- ▶ That is
Average productivity of fishing = cost of fishing
- ▶ But social welfare is

$$W(N) = F(N) - cN.$$

- ▶ Thus the optimum is

$$F'(N) = c.$$

- ▶ That is
Marginal productivity of fishing = cost of fishing.
- ▶ The average/marginal distinction is the essence of a congestion externality.

1. Fishing Clubs

Subponds

- ▶ The pond is cut into K identical subponds
- ▶ If n people fish in a subpond, they catch $F(Kn)/K$ fish
- ▶ if N fishermen are uniformly distributed over the subponds, they catch $KF(Kn)/K = F(N)$ fish
- ▶ Each subpond is owned by a firm which charges an entry fee p per fisherman.
- ▶ These firms compete for customers but earn positive rents due to their ownership of part of the ponds
- ▶ Competition brings their prices in line, but a firm can charge a higher price and remain competitive if in exchange for that it offers less congestion, i.e. more fish, to its clients (members)

1. Fishing Clubs

Subponds

- ▶ The value to a customer of a pond which charges p and has n people is

$$v(p, n) = \frac{F(Kn)}{Kn} - p - c.$$

- ▶ In equilibrium, all subponds such that $n > 0$ (called "active") must yield the same value to the customers :

$$\frac{F(Kn)}{Kn} - p - c = \bar{u}.$$

- ▶ This generate an equilibrium relationship between the density of fishermen at an active pond and its entry fee :

$$p(n) = \frac{F(Kn)}{Kn} - c - \bar{u}. \quad (1)$$

1. Fishing Clubs

Profit maximization

- ▶ Since owners maximize profits, they will choose the density that maximizes $np(n)$:

$$n = \arg \max np(n) = n \left(\frac{F(Kn)}{Kn} - c - \bar{u} \right).$$

- ▶ The FOC of firms is $F'(Kn) = c + \bar{u}$.
- ▶ At a symmetrical equilibrium, $n = N/K$ and therefore : $F'(N) = c + \bar{u}$.
- ▶ Since there are many (many) potential fishermen that can freely join clubs, in equilibrium $\bar{u} = 0$.
- ▶ Therefore, $F'(N) = c$.
- ▶ The equilibrium with competitive clubs is efficient.

1. Fishing Clubs

Efficient equilibrium

- ▶ The price schedule defined by (1) accurately reflects the congestion externality : clubs with more fishermen must charge a lower price.
- ▶ This makes the total profit of a club equal to the economic surplus it creates, which leads to an efficient outcome.
- ▶ The theory of competitive search (Moene (1995)) applies these ideas to the Mortensen-Pissarides matching model.

2. Search Clubs

Assumptions

- ▶ There are several ways of doing it. Here we assume "wage posting".
- ▶ firms and workers can join a search intermediary, i.e. they *assign* their vacancy and candidacy to one of those clubs.
- ▶ Vacancies and unemployed workers are only matched together if they belong to the same club.
- ▶ Each club specifies the wage that must be paid to the worker after the match has taken place.
- ▶ A club contains vacancies and unemployed workers in proportions θ , and specifies the wage w that will be paid after a vacancy is matched with an unemployed.
- ▶ The unemployed members of the club are charged f per unit of time until they are matched, while vacancies are charged g per unit of time.

2. Search Clubs

Values

- ▶ In equilibrium, the value of being unemployed V_u which is the maximum value offered to the unemployed across clubs.
- ▶ V_v is the maximum of posting a vacancy across clubs. With free entry, $V_v = 0$.
- ▶ Let $V_u(w, \theta, f)$ the value to the unemployed of joining a club with (w, θ, f) . The asset valuation equation is

$$rV_u(w, \theta, f) = -f + \theta q(\theta)(V_e(w) - V_u(w, \theta, f)),$$

where $V_e(w)$ is the value of a job to the worker paying a wage w , which satisfies

$$rV_e(w) = w + s(V_u - V_e(w)).$$

- ▶ Note : We restrict here to stationary equilibrium, so that all $\dot{V}$ are 0.

2. Search Clubs

Equilibrium

- ▶ In equilibrium, any club who is able to attract workers must be such that $V_u(w, \theta, f) \geq V_u$.
- ▶ But since clubs maximize profits, they will not want to offer the unemployed a utility greater than V_u , which is the maximum offered by other clubs.
- ▶ This determines the fee charged by a club which picks w and θ :

$$rV_u(w, \theta, f) = -f + \theta q(\theta)(V_e(w) - V_u(w, \theta, f)),$$

$$rV_u = -f + \theta q(\theta)(V_e(w) - V_u(w, \theta, f)),$$

$$f(w, \theta) = \theta q(\theta)V_e(w) - (r + \theta q(\theta))V_u. \quad (2)$$

2. Search Clubs

Equilibrium (continued)

- ▶ Value of posting a vacancy at a club with (w, θ, g) is $V_v(w, \theta, g)$ such that

$$rV_v(w, \theta, g) = -c - g + q(\theta)(J(w) - V_v(w, \theta, g)),$$

- ▶ where $J(w)$ is the value to the firm of a position with wage w , i.e.

$$rJ(w) = y - w - sJ(w).$$

- ▶ In equilibrium it must be that $V_v(w, \theta, g) \geq 0$
- ▶ Profit maximization by clubs will imply $V_v(w, \theta, g) = 0$,
- ▶ Therefore $V_v(w, \theta, g) = 0$, implying

$$g(w, \theta) = -c + q(\theta)J(w). \quad (3)$$

2. Search Clubs

Equilibrium (continued)

- ▶ Let's look at competition between clubs
- ▶ The profit of a club, per unemployed member, is given by $\pi(w, \theta) = f(w, \theta) + \theta g(w, \theta)$.
- ▶ Free entry of clubs imply that
 - ▶ This cannot exceed zero in equilibrium, for *any* potential club offering w and θ . Therefore :

$$f(w, \theta) + \theta g(w, \theta) \leq 0, \forall w, \theta.$$

- ▶ Profits must be strictly zero for clubs that are nonempty in equilibrium.

2. Search Clubs

Equilibrium (continued)

- Compute $\pi(w, \theta)$ using (2) and (3)

$$f(w, \theta) = \theta q(\theta) V_e(w) - (r + \theta q(\theta)) V_u \quad (2)$$

$$g(w, \theta) = -c + q(\theta) J(w) \quad (3)$$

- to get

$$\pi(w, \theta) = \theta q(\theta) (V_e(w) + J(w) - V_u) - (c\theta + rV_u).$$

- $\theta q(\theta) (V_e(w) + J(w) - V_u)$: benefit of the club, per unemployed member, in terms of job creation.
- $(c\theta + rV_u)$: cost of search per unemployed member.
- $V_e(w) + J(w) - V_u =$ Total net surplus of a match
- $rV_e(w) = w + s(V_u - V_e(w))$ implies $V_e = \frac{w+sV_u}{r+s}$
- $rJ(w) = y - w - sJ(w)$ implies $J = \frac{y-w}{r+s}$
- Therefore, $V_e(w) + J(w) - V_u = \frac{y-rV_u}{r+s}$

2. Search Clubs

Equilibrium (continued)

- ▶ Since $V_e(w) + J(w) - V_u = \frac{y-rV_u}{r+s}$, $\pi(w, \theta) = \pi(\theta)$ does not depend on w .
 - ▶ This is due to the fact that w is a transfer between the firm and the worker, which leaves the total surplus of the match unchanged.
 - ▶ Any increase in w would be matched by an increase in f and a fall in g to maintain V_u and V_v unchanged.

2. Search Clubs

Equilibrium (continued)

- ▶ The preceding equations imply that the equilibrium value of θ , θ^{cs} satisfies

$$\theta^{cs} = \arg \max \pi(\theta) = \theta q(\theta) \frac{y - rV_u}{r + s} - c\theta - rV_u.$$

- ▶ Furthermore,

$$\pi(\theta^{cs}) = 0.$$

- ▶ These two equations jointly determine θ_E and V_u in equilibrium.
- ▶ **Note :** There is only one type of club in equilibrium.
- ▶ Working through those equations yields the equilibrium θ^{cs} , which must satisfy

$$y(q(\theta^{cs}) + \theta^{cs} q'(\theta^{cs})) = c(r + s - (\theta^{cs})^2 q'(\theta^{cs})). \quad (4)$$

3. Comparing equilibrium and optimal value of θ

θ^{cs} and θ^{so}

- ▶ Competitive Search Equilibrium θ^{cs} is given by

$$y(q(\theta^{cs}) + \theta^{cs} q'(\theta^{cs})) = c(r + s - (\theta^{cs})^2 q'(\theta^{cs})). \quad (4)$$

- ▶ Socially optimal one θ^{so} ?
- ▶ Recall from Lecture 10 that Social Planner FOC are

$$c = \lambda_t \frac{\partial m(u, v)}{\partial v} \quad (5)$$

$$r\lambda_t = y - s\lambda_t - \lambda_t \frac{\partial m}{\partial u} + \dot{\lambda}_t \quad (6)$$

3. Comparing equilibrium and optimal value of θ

Computing θ^{so}

- ▶ We can re-express those conditions in terms of θ .
- ▶ Note that

$$m(u, v) = u\theta q(\theta).$$

- ▶ Therefore,

$$\begin{aligned}\frac{\partial m(u, v)}{\partial v} &= u \times [q(\theta) + \theta q'(\theta)] \frac{\partial \theta}{\partial v} \\ &= q(\theta) + \theta q'(\theta).\end{aligned}$$

- ▶ Hence

$$\lambda_t = \frac{c}{q(\theta_t) + \theta_t q'(\theta_t)}. \quad (7)$$

- ▶ Also,

$$\begin{aligned}\frac{\partial m(u, v)}{\partial u} &= \theta q(\theta) + u[q(\theta) + \theta q'(\theta)] \frac{\partial \theta}{\partial u} \\ &= \theta q(\theta) - \theta[q(\theta) + \theta q'(\theta)] \\ &= -\theta^2 q'(\theta).\end{aligned} \quad (8)$$

3. Comparing equilibrium and optimal value of θ

Computing θ^{so} (continued)

- ▶ Substituting (7) and (8) into (6) we get the evolution of θ^{so} :

$$\begin{aligned} \frac{rc}{q(\theta) + \theta q'(\theta)} &= y - \frac{sc}{q(\theta) + \theta q'(\theta)} + \frac{c\theta^2 q'(\theta)}{q(\theta) + \theta q'(\theta)} \\ &+ \frac{d}{dt} \left[\frac{c}{q(\theta_t) + \theta_t q'(\theta_t)} \right]. \end{aligned}$$

- ▶ This equation only involves θ and is unstable. Therefore the only non explosive solution is the stationary one, i.e.

$$(r + s - \theta^2 q'(\theta))c = (q(\theta) + \theta q'(\theta)) y. \quad (9)$$

- ▶ This determines the optimal $\theta = \theta^{so}$, which is constant over time, like the equilibrium one.

3. Comparing equilibrium and optimal value of θ

$$\theta^{cs} = \theta^{so}$$

- Competitive search :

$$y(q(\theta^{cs}) + \theta^{cs} q'(\theta^{cs})) = c(r + s - (\theta^{cs})^2 q'(\theta^{cs})). \quad (4)$$

- Optimum :

$$(r + s - (\theta^{so})^2 q'(\theta^{so}))c = (q(\theta^{so}) + \theta^{so} q'(\theta^{so})) y. \quad (9)$$